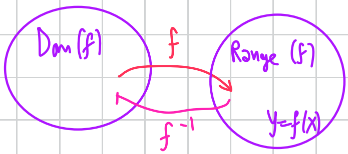
Inverse functions

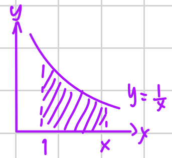
f is 1-1 : if $x_1, x_2 \in Dom(f)$, $x_1 \neq x_2$ then $f(x_1) \neq f(x_2)$



If f is 1-1, define $f^{-1}(y) = x$; $x \in Dom(f)$ s.t. $f(x) = y$

1. If f contin, then so in f^{-1}
2. If f is differentiable at $x=a$, then f^{-1} is differentiable at $x = f(a) = b$

$$(f^{-1}(b))' = \frac{1}{f'(a)} \quad \text{, provided } f'(a) \neq 0$$

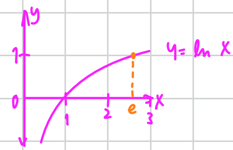


7.2 Natural logarithms (ln)

Def $\forall x > 0$, define $\ln x = \int_1^x \frac{1}{t}\, dt$

Remarks : 1) $\ln(1) = 0$

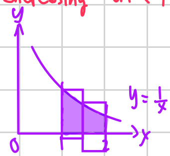
2) $\ln x \begin{cases} \oplus & \text{if } x > 1 \\ \ominus & \text{if } 0 < x < 1 \end{cases}$

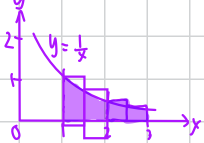


3) $\frac{d}{dx} \ln x = \frac{1}{x}$ (By F.T.C.) > 0 (Increasing on $(0, \infty)$)

$\frac{d^2}{dx^2} \ln x = -\frac{1}{x^2} \rightarrow$ Concave down

4) $\exists e$ s.t. $\ln e = 1$ $(2 < e < 3)$





Strictly increasing
→ exactly 1 e

Pf: $\ln 2 = \int_1^2 \frac{1}{x} dx < (1 + 0.5) \times 0.5$
< 1

$\ln 3 = \int_1^3 \frac{1}{x} dx > (1 + 0.5 + 0.4 + 0.33) \times 0.5 = 1.116 > 1$

IVT, there exist e such that $\ln(e) = 1$ $\qquad e = 2.718\ldots$

Laws of ln

1) $\ln(ab) = \ln a + \ln b$

2) $\ln\left(\frac{1}{a}\right) = -\ln a$; $a > 0$

Proof $\ln \frac{1}{n} : \int_1^{1/n} \frac{1}{x}\, dx$ $\qquad$ let $u = 1/x \rightarrow du = -\frac{1}{x^2} dx$

$= \int_1^n u \frac{du}{(-u^2)}$

$= \int_1^n -\frac{1}{u}\, du$

$= -\ln n$

3) $\ln\left(\frac{a}{b}\right) = \ln\left(a \frac{1}{b}\right) = \ln a + \ln\left(\frac{1}{b}\right) = \ln a - \ln b$

4) $\ln(a^b) = b \ln a$

Proof: $f(x) = \ln x^b - b \ln x$, $x > 0$

$f(1) = 0$, $f'(x) = \frac{1}{x^b} \cdot bx^{b-1} - \frac{b}{x} \equiv 0 \quad \forall x > 0$

$\therefore f(x) \equiv f(1) = 0$ $\quad$ Take $x = a \rightarrow \ln(a^b) - b \ln a = 0$

Remarks (more)

5) $\ln x > \ln 2^n \overset{\text{law 4}}{=} n\ln 2 \to \infty$ when $n\to\infty$ (if $x > 2^n$)

$\therefore \lim_{x\to\infty} \ln x = +\infty$ "$\ln \infty = \infty$", "$\ln 0^+ = -\infty$"

6) $\lim_{x\to 0^+} \ln x \xrightarrow{\text{law 2}} \lim_{x\to 0^+} -\ln \frac{1}{x}$

$u=\frac{1}{x}$ $\lim_{u\to\infty} -\ln u \overset{\text{Rk 5}}{=} -\ln\infty = -\infty$

$x=0 \to$ vertical asymptote of $\ln x$

$\int \frac{1}{x}\,dx = \ln|x| + C$

Recall $(\ln x)' = \frac{1}{x}$ $(x>0)$

$(\ln|x|)' \xrightarrow{x\neq 0} \frac{1}{|x|}|x|'$

$= \begin{cases} \frac{1}{|x|} ; & x>0 \\ -\frac{1}{|x|} ; & x<0 \end{cases} \overset{\text{"}}{=} \frac{1}{x}$

Application of $\ln x$

1. $\int \tan x\,dx = \int \frac{\sin x}{\cos x}\,dx = \int -\frac{1}{u}\,du = -\ln x + C = -\ln(\cos x) + C$ ($u=\cos x$, $du = -\sin x\,dx$)

2. $\int \sec x\,dx = \int \frac{1}{\cos x}\,dx = \int \frac{\cos x}{\cos^2 x}\,dx = \int \frac{\cos x}{1-\sin^2 x}\,dx = \int \frac{1}{1-\sin^2 x}\,(d(\sin x))$

$= \int \frac{1}{1-u^2}\,du$

$= \int \frac{1}{2}\left(\frac{1}{1-u} + \frac{1}{1+u}\right) du$

$= \frac{1}{2}\left(\int \frac{1}{1-u}\,du + \int \frac{1}{1+u}\,du\right)$

$w = 1-u \to dw = -du$, $v = 1+u \to dv = du$

$= \frac{1}{2}\left(\int -\frac{1}{w}\,dw + \int \frac{1}{v}\,dv\right)$

$= \frac{1}{2}\left(-\ln|w| + \ln|v|\right) + C$

$= \frac{1}{2}\ln\left|\frac{1+\sin x}{1-\sin x}\right|\frac{\times(1+\sin x)}{\times(1+\sin x)} + C$

$= \frac{1}{2}\ln\left|\frac{(1+\sin x)^2}{\cos^2 x}\right| + C$

$= \frac{1}{2}\left(2\ln\left|\frac{1+\sin x}{\cos x}\right|\right) + C$

$= \ln|\sec x + \tan x| + C$

3. logarithmic functions

Given $y = \text{ugly}(x)$ want $\frac{dy}{dx} = ?$

Strategy: $\ln y = \ln(\text{ugly}(x))$

$\frac{d}{dx}\ln y = \frac{d}{dx}\ln(\text{ugly}(x))$

$\frac{1}{y}\frac{dy}{dx} = \frac{d}{dx}(\ln(\text{ugly}(x)))$

$\frac{dy}{dx} = y \cdot \frac{d}{dx}\ln(\text{ugly}(x))$

7.2/64 $y = \frac{\theta\sin\theta}{\sqrt{\sec\theta}}$ $\frac{dy}{d\theta} = ?$

$\ln y = \ln\left(\frac{\theta\sin\theta}{\sec\theta}\right) = \ln\theta + \ln(\sin\theta) - \ln(\sqrt{\sec\theta})$

$= \ln\theta + \ln(\sin\theta) + \frac{1}{2}\ln(\cos\theta)$

$\frac{d}{d\theta}\ln y = \frac{d}{d\theta}(\ln\theta + \ln(\sin\theta) + \frac{1}{2}\ln(\cos\theta))$

$\frac{1}{y}\frac{dy}{d\theta} = \frac{1}{\theta} + \frac{\cos\theta}{\sin\theta} - \frac{1}{2}\frac{\sin\theta}{\cos\theta}$

$\frac{dy}{d\theta} = y\left(\frac{1}{\theta} + \cot\theta - \frac{1}{2}\tan\theta\right)$

7.3 Exponential Functions

Let $y = f(x) = \ln x,\ x > 0$

Consider $f^{-1}(x)$: $f^{-1}(x)$ is denoted as e^x $\left(\begin{matrix}\mathrm{Dom}(e^x): (-\infty,\infty) \\ \mathrm{Range}(e^x): (0,\infty)\end{matrix},\ \begin{matrix}\mathrm{Dom}(\ln x) = (0,\infty) \\ \mathrm{Range}(\ln x) = (-\infty,\infty)\end{matrix}\right)$

Properties of e^x: 1. $\ln(e^x) = x \quad \forall x \in (-\infty,\infty)$

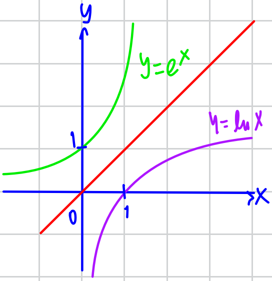
$(f(f^{-1}(x)) = x \quad \forall x \in \mathrm{Dom}(f^{-1}))$

2. $e^{\ln x} = x \quad \forall x \in (0,\infty)$

$(f^{-1}(f(x)) = x \quad \forall x \in \mathrm{Dom}(f))$

3. $(e^y)' = (f^{-1}(y))' = \dfrac{1}{f'(x)} = \dfrac{1}{\frac{1}{x}} = x = e^y \quad \therefore \boxed{\dfrac{de^y}{dy} = e^y \quad \forall y \in (-\infty,\infty)}$

$(y = \ln x \to x = e^y)$



4. $e^0 = 1$

5. $\lim\limits_{x\to\infty} e^x = \infty, \quad \lim\limits_{x\to-\infty} e^x = 0$

6. $\int e^x\,dx = e^x + C$

$\left(a^{x_1+x_2} = e^{(x_1+x_2)\ln a} = e^{x_1\ln a + x_2 \ln a} = e^{x_1 \ln a}\cdot e^{x_2\ln a} = a^{x_1}\cdot a^{x_2}\right)$

Law of exponents: 1) $e^{x_1+x_2} = e^{x_1}\cdot e^{x_2}$ $\quad \to a^{x_1+x_2} = a^{x_1}\cdot a^{x_2}$

Proof: $\ln(e^{x_1+x_2}) = x_1 + x_2 = \ln(e^{x_1}) + \ln(e^{x_2}) = \ln(e^{x_1}\cdot e^{x_2})$

2) $e^{-x_1} = \dfrac{1}{e^{x_1}}$ $\quad \to a^{-x} = \dfrac{1}{a^x}$

3) $\dfrac{e^{x_1}}{e^{x_2}} = e^{x_1 - x_2}$ $\quad \to \dfrac{a^{x_1}}{a^{x_2}} = a^{x_1-x_2}$

4) $(e^{x_1})^{x_2} = e^{x_1x_2}$ $\quad \to (a^{x_1})^{x_2} = a^{x_1x_2}$

General exponential functions

Let $a > 0$ const : $a^x = e^{x\ln a}$ $\quad (x \in (-\infty,\infty))$

Properties 1) $a^0 = e^0 = 1$

2) $a^x > 0$

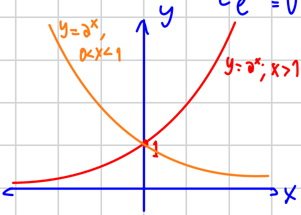
3) $(a^x)' = e^{x\ln a}\cdot \ln a = a^x \cdot \ln a = \begin{cases} \oplus & ; a>1 \\ 0 & ; a=1 \\ \ominus & ; a<1 \end{cases}$, $a''(x) = a^x (\ln a)^2 > 0$ if $a>0, a\neq 1$

$\to a^x$ is concave up

4) $\int a^x\,dx + C = \dfrac{a^x}{\ln a} + C \ ; a>0,\ a\neq 1$

จำนวนลบ ยกกำลังได้แค่บางตัว $((-2)^{\sqrt{2}}$ undefined, $(-2)^{1/3}, (-2)^{9/29}$ defined$)$

$\lim\limits_{x\to+\infty} a^x = e^{+\infty\cdot\ln a} = \begin{cases} \infty & \text{if } a>1 \\ e^{-\infty} = 0 & \text{if } a<1 \end{cases}$, $\quad \lim\limits_{x\to-\infty} a^x = e^{-\infty\cdot\ln a} \begin{cases} 0 & , a>1 \\ \infty & , a<1 \end{cases}$



Logarithms with general base $a>0$, $a\neq 1$

Let $y=f(x)=a^x$, $\text{Dom}(f)=(-\infty,\infty) \to \text{Range}(f)=(0,\infty)$

And f is $\nearrow$ $(a>1)$ or $\searrow$ $(a<1)$ $\therefore$ f is 1-1 $\to f^{-1}$ exists

$\text{Dom}(f^{-1})=\text{Range}(f)=(0,\infty)$, $\text{Range}(f^{-1})=\text{Dom}(f)=(-\infty,\infty)$

Let $\log_a x$ to be $f^{-1}(x)$

$$\log_a(a^x)=x, \quad \forall x\in(-\infty,\infty) \qquad a^{\log_a x}=x \quad \forall x\in(0,\infty)$$

$$a^{\log_a x}=x$$

$$\log_a x\cdot \ln a=\ln x \to \boxed{\log_a x=\frac{\ln x}{\ln a}}$$

$$\boxed{(\log_a x)'=\left(\frac{\ln x}{\ln a}\right)'=\frac{1}{x\ln a}} \qquad \boxed{\frac{d}{dx}(\log_a u)=\frac{1}{\ln a}\cdot\frac{1}{u}\frac{du}{dx}}$$

e.g. $\dfrac{d(\sin x)^{\cos x}}{d/}$; $y=\sin x^{\cos x}$

$$\frac{d}{dx}\ln y=\frac{d}{dx}\ln(\sin x)^{\cos x}$$

$$\frac{1}{y}\frac{dy}{dx}=\frac{d}{dx}(\cos x\cdot\ln(\sin x))=-\sin x\ln\sin x+\cot x\cdot\cos x$$

$$\frac{dy}{dx}=\sin x\cos x=\sin x^{\cos x}\left(-\sin x\ln(\sin x)+\frac{\cos^2 x}{\sin x}\right)$$

This will be in the quiz ☺

7.5 Indeterminate forms & L'Hopital's rule

↳ $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0\cdot\infty$, 1^∞, $\infty-\infty$, ∞^0, 0^0

$(0.5)^\infty=0$

$(1.1)^\infty=\infty$

L'Hopital's rule: let f & g be differentiable on $[a,b]$ and $g'(x)\neq 0$ on $[a,b]$

Assume $\lim\limits_{x\to a^+}\dfrac{f'(x)}{g'(x)}=l$ (maybe $\pm\infty$)

1) If $\lim\limits_{x\to a^+}f(x)=0=\lim\limits_{x\to a^+}g(x)$, then $\lim\limits_{x\to a^+}\dfrac{f(x)}{g(x)}$ exists and equals $\lim\limits_{x\to a^+}\dfrac{f'(x)}{g'(x)}=l$ ($\frac{0}{0}$ Type) (allow to be $\pm\infty$)

2) If $\lim\limits_{x\to a^+}g(x)=\pm\infty$, then $\lim\limits_{x\to a^+}\dfrac{f(x)}{g(x)}$ (Type $\frac{??}{\infty}$) exists and equals $\lim\limits_{x\to a^+}\dfrac{f'(a)}{f'(b)}$ Hard (Prove in MAT 2050)

Cauchy's Mean Value Theorem

Let f & g be contin on $[a,b]$ & differentiable on (a,b), then $\exists c\in(a,b)$ s.t.

$$\frac{f(b)-f(a)}{g(b)-g(a)}=\frac{f'(c)}{g'(c)}$$

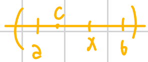
Proof Let $h(x)=(f(b)-f(a))g(x)-(g(b)-g(a))f(x)$

- h contin of $[a,b]$, h diff on (a,b)
- $h(a)=f(b)g(a)-f(a)g(b)$
- $h(b)=f(b)g(a)-f(a)g(b)$

$\xrightarrow{\text{Rolle's}}$ There exists $c\in[a,b]$ s.t. $h'(c)=0$ QED

Proof of L'Hopital's :

extend f, g s.t. $f(a) = 0 = g(a) \rightarrow f$ & g contin on $[a, b)$



$\forall x \in (a, b)$ fixed, $f(c)$ is contin on $[a, x]$, diff on (a, x)

Now Cauchy's MVT applies $\longrightarrow \dfrac{f(x) - f(a)}{g(x) - g(a)} = \dfrac{f'(c)}{g'(c)}$ $\exists c \in (a, x)$

$x \to a^+ \therefore c \to a^+$

$$\therefore \lim_{x \to a^+} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{c \to a^+} \frac{f'(c)}{g'(c)} = \ell$$

$(g' \neq 0$ on $[a, b])$

e.g. 1 $\displaystyle\lim_{x \to 0} \frac{\sin x - x}{x^3}$ $\quad \frac{0}{0}$

$$\overset{L'H}{=} \lim_{x \to 0} \frac{\cos x - 1}{3x^2} \overset{L'H}{=} \lim_{x \to 0} \frac{-\sin x}{6x} = \frac{-1}{6}$$

$\frac{\sin x}{x} \to 1$

e.g. 2 $\displaystyle\lim_{x \to \infty} \frac{\log_a x}{b^x}$; $a > 0, a \neq 1, b > 1$

$$= \lim_{x \to \infty} \frac{\frac{\ln x}{\ln a}}{b^x} \overset{L'H}{=} \lim_{x \to \infty} \frac{\frac{1}{x \ln a}}{b^x \cdot \ln b} = \frac{1}{\ln a \cdot \ln b} \cdot \lim_{x \to \infty} \frac{1}{\underset{\to\infty}{x} \cdot \underset{\to\infty}{b^x}} = 0$$

Moral of the story : $b(x) > \log_a x$ as $x \to \infty$

e.g. 3 $\displaystyle\lim_{x \to \infty} \frac{x^a}{b^x}$ $\left(\frac{\infty}{\infty}\right)$ $\displaystyle\overset{L'H}{=} \lim_{x \to \infty} \frac{a x^{a-1}}{b^x \ln b}$ $\overset{\text{L'H หลายรอบ}}{=}$ $\displaystyle\lim_{x \to \infty} \frac{a(a-1)\ldots(1)}{b^x (\ln b)^a} = \frac{a!}{(\ln b)^a} \cdot \lim_{x \to \infty} \frac{1}{b^x} = 0$

e.g. 4 $\displaystyle\lim_{x \to 0^+} x^x$ $\quad (0^0$ type$)$

Let $y = x^x \rightarrow \ln y = x \ln x$

$$\lim_{x \to 0^+} \ln y = \lim_{x \to 0^+} \underset{0 \cdot (-\infty)}{x \ln x} = \lim_{x \to 0^+} \frac{\ln x}{\frac{1}{x}} = \lim_{x \to 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = \lim_{x \to 0^+} -x = 0$$

$$\therefore \lim_{x \to 0^+} x^x = \lim_{x \to 0^+} y = \lim_{x \to 0^+} e^{\ln y} = e^{\lim_{x \to 0^+} \ln y} = e^{\lim_{x \to 0^+} x \ln x} = e^0 = 1$$

e.g. 5 $\displaystyle\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x$ $\quad 1^\infty$

$$y = \left(1 + \frac{1}{x}\right)^x \rightarrow \ln y = x \ln\left(1 + \frac{1}{x}\right)$$

$$\lim_{x \to \infty} y = \lim_{x \to \infty} \frac{\ln\left(1 + \frac{1}{x}\right)}{\frac{1}{x}} = \lim_{x \to \infty} \frac{\frac{1}{1 + \frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right)}{-\frac{1}{x^2}} = \lim_{x \to \infty} \frac{-\frac{1}{x \cdot (x+1)}}{-\frac{1}{x^2}} = \lim_{x \to \infty} \frac{x}{x+1} = 1$$

$$\therefore \lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = \lim_{x \to \infty} y = \lim_{x \to \infty} e^{\ln y} = e^1 = e$$

7.8 Relative Rates of Growth & Decay

Concern: Behavior of $f(x)$ as $x \to c$ (may be $\pm\infty$)

Recall $f(x) = O(1)$ as $x \to c$ $\quad \exists$ const a s.t. $|f(x)| \leq a$ for $x \approx c, x \neq c \rightarrow f$ is bounded

$f(x) = o(1)$ as $x \to c \iff \displaystyle\lim_{x \to 0} f(x) = 0$

e.g. $\cos x, \sin x = O(1)$ as $x \to c$, $\dfrac{1}{1 + x^2} = o(1)$ as $x \to \infty$

$\frac{\sin x}{x} = O(1)$ as $x \to 0$ $\left(\lim_{x\to 0} \frac{\sin x}{x} = 1\right)$

Def We say f(x) is of at most the order of g as $x \to c$ if $f(x) = g(x)O(1)$, $x \approx c$, $x \neq c$

e.g. $f(x) = \underset{g}{x} \underbrace{\sin x}_{O(1)}$ is at most order of x

e.g. $\tan(x) = \frac{\sin x}{\cos x} = \sin x \sec x$ As $x \to \frac{\pi}{2}$
$= O(1)\sec x$

$\therefore \tan x$ is at most order of $\sec x$

f is of smaller order of g if $f(x) = o(1)\, g(x) = o(g(x))$

Facts 1) $O(1) + O(1) = O(1)$
2) $O(1)O(1) = O(1)$
3) $o(1) + o(1) = o(1)$
4) $o(1)\, o(1) = o(1)$
5) If $\lim_{x\to c} f(x)$ exists as a finite number, then $f(x) = O(1)$ as $x \to c$
6) $o(1) \cdot O(1) = o(1)$ by squeeze thm.
7) $o(1) = O(1)$

Notes: 1) if $f(x) > 0$, $g(x) > 0$, then $f(x) = o(g(x)) \Longleftrightarrow \lim_{x\to c} \frac{f(x)}{g(x)} = 0 \Leftrightarrow \lim_{x\to c} \frac{g(x)}{f(x)} = \infty$

g(x) grows faster than f(x)

Moreover, if $\lim_{x\to c} f(x) = 0 = \lim_{x\to c} g(x)$, we say f decays faster than g
(Both infinitesimal) or f is higher order infinitesimal than g

2) If $f(x) > 0$, $g(x) > 0$ and $\lim_{x\to c} \frac{f(x)}{g(x)} = \text{finite } l > 0$ then we say f & g grow/decay in the same rate

3) $b^x >> x^a >> \log_a x$; $x \to \infty$, $b > 1$, $a > 0$ $(a \neq 1)$

4) Algebraic decay vs exponential decay

$\frac{1}{x^a}$ vs b^{-x} as $x \to \infty$, $b > 1$

$$\lim_{x\to\infty} \frac{\frac{1}{x^a}}{b^{-x}} = \lim_{x\to\infty} \frac{b^x}{x^a} \overset{L'H}{=} +\infty$$

(b^{-x} decays faster $\to$ higher order infinitesimal)